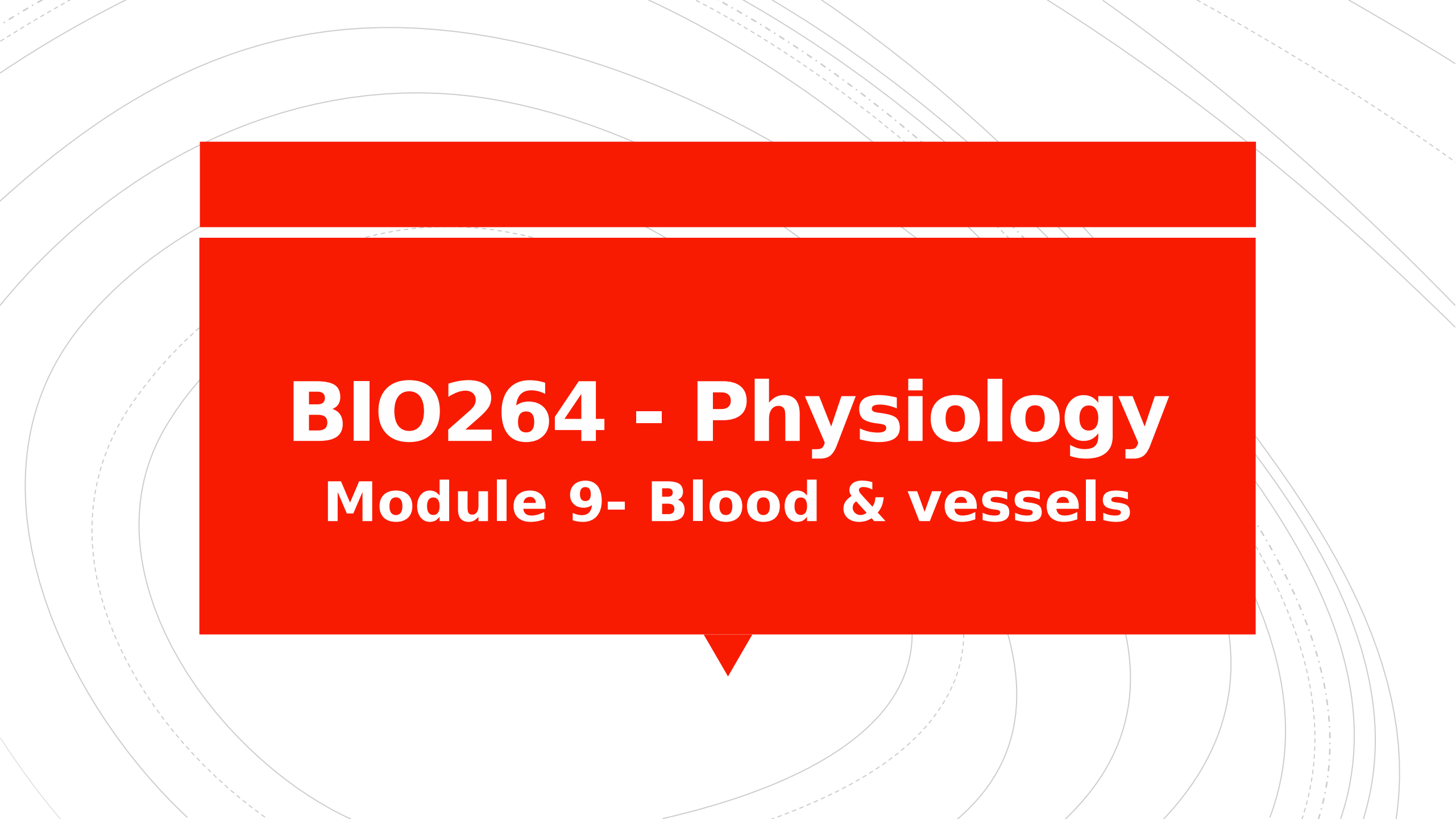
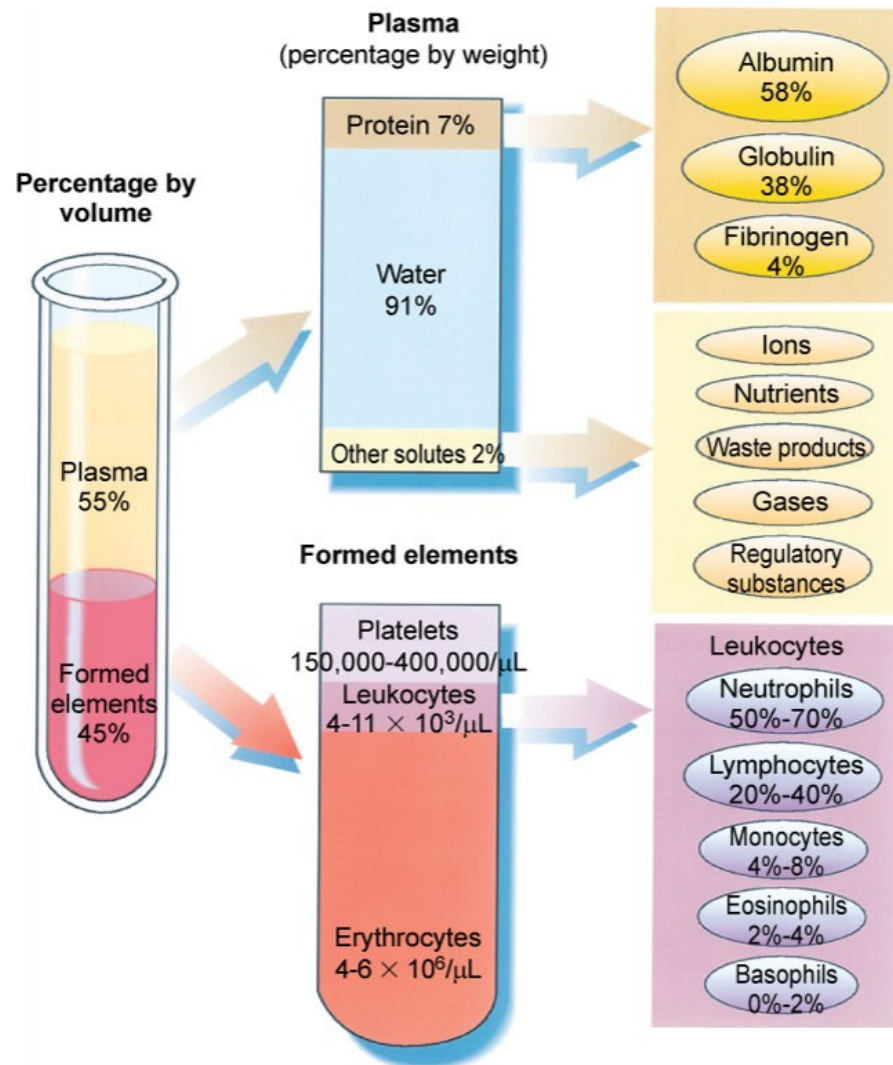


The background features a series of concentric circles in light gray, some solid and some dashed, creating a ripple effect. A large red speech bubble is centered on the page, containing the text.

BIO264 - Physiology

Module 9- Blood & vessels

BLOOD



From Thibodeau GA, Patton KT: *Anatomy and physiology*, ed 6, St Louis, 2007, Mosby.

❖ Fluid Connective tissue

❖ Functions:

❖ **Transportation-**

O₂, nutrients, hormones, waste

❖ **Regulation-**

fluid/electrolytes, acid/base balance, **body temp through heat distribution**

❖ **Protection-**

coagulation (prevents fluid loss), infections

pH?

Why is it important to maintain normal blood pH?

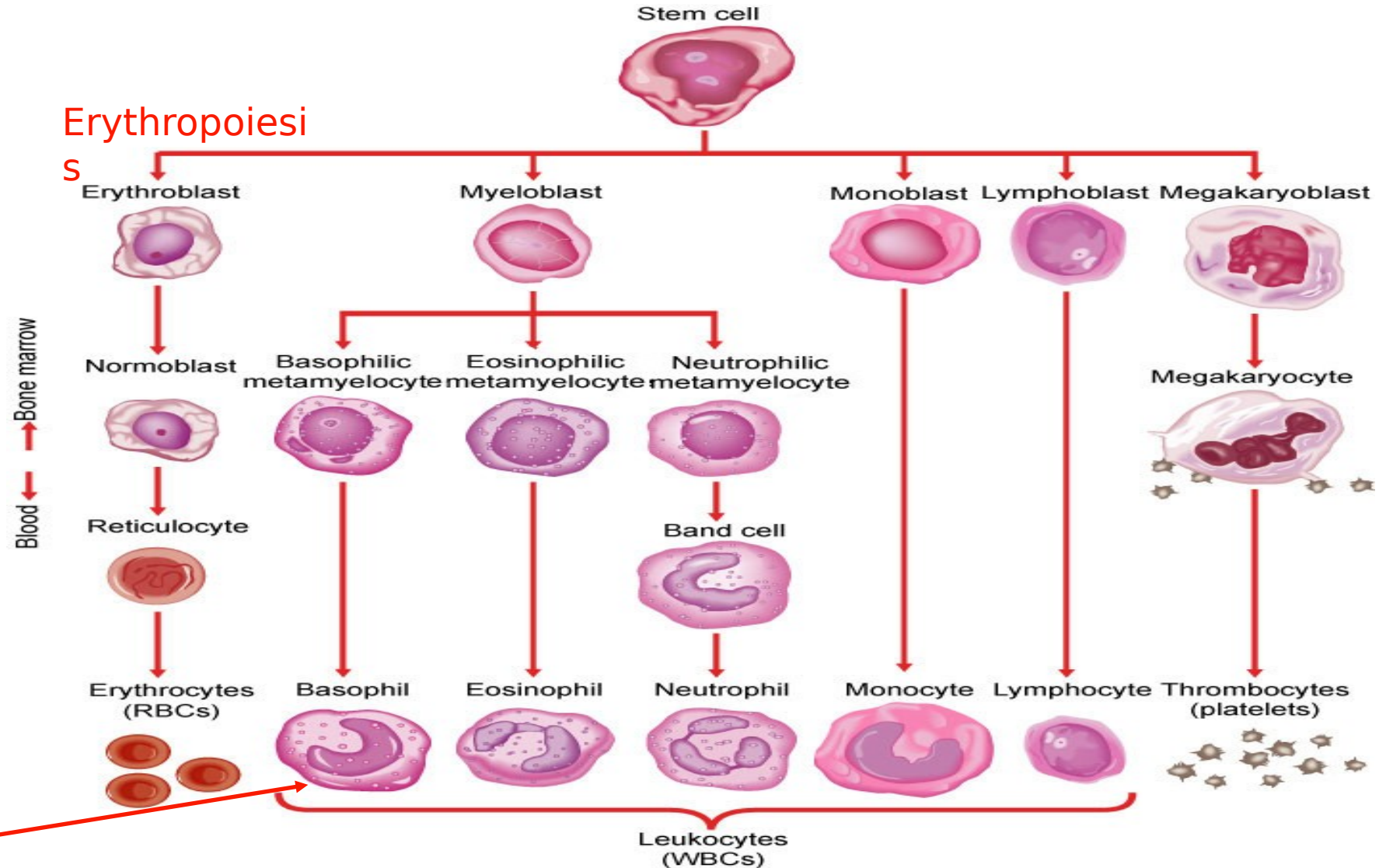
How are they all made? HEMATOPOIESIS

RBC/erythrocytes
lives 120 days,
carries O₂

Platelets: cell
fragment, lives
9-12 days,
clotting!

Granulocytes:
Basophil
Eosinophil
Neutrophil

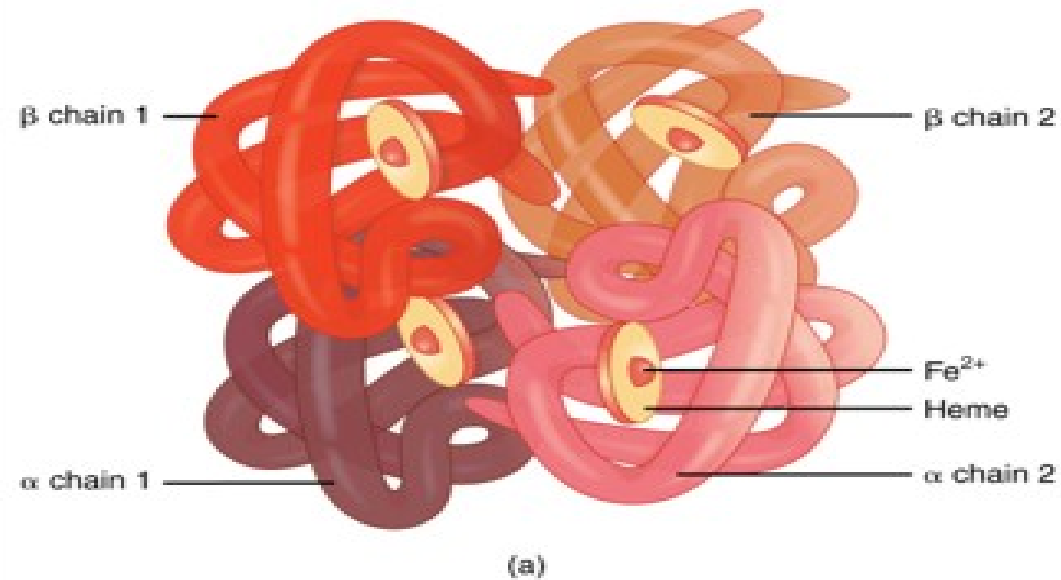
Phil has
sand(grains) in
his shoes



Mosby items and derived items © 2011, 2007 by Mosby, Inc., an affiliate of Elsevier, Inc.

Leukocytes defend against pathogens

Hemoglobin



- Quarternary protein
- Water soluble
- Binds ALL gasses: O_2 , CO_2 (as carbonic acid), CO (poisonous)
- ~300 /RBC, 5-6 million RBCs – so we have the ability to carry about 18 billion O_2 molecules in our blood

Practice Question

- Which of the following regarding red blood cells is **FALSE**?
 - A – Once produced, they spend approximately 120 days carrying out their function
 - B – Once their lifespan is up, the hemoglobin is broken down and the amino acids are recycled
 - C – Once their lifespan is up, the macrophages of the brain engulf them to begin the breakdown process
 - D – Bilirubin indirectly give urine and feces their color

White Blood Cell Characteristic s -matching

1. Diapedesis

A. Change shape

2. Amoeboid
Movement

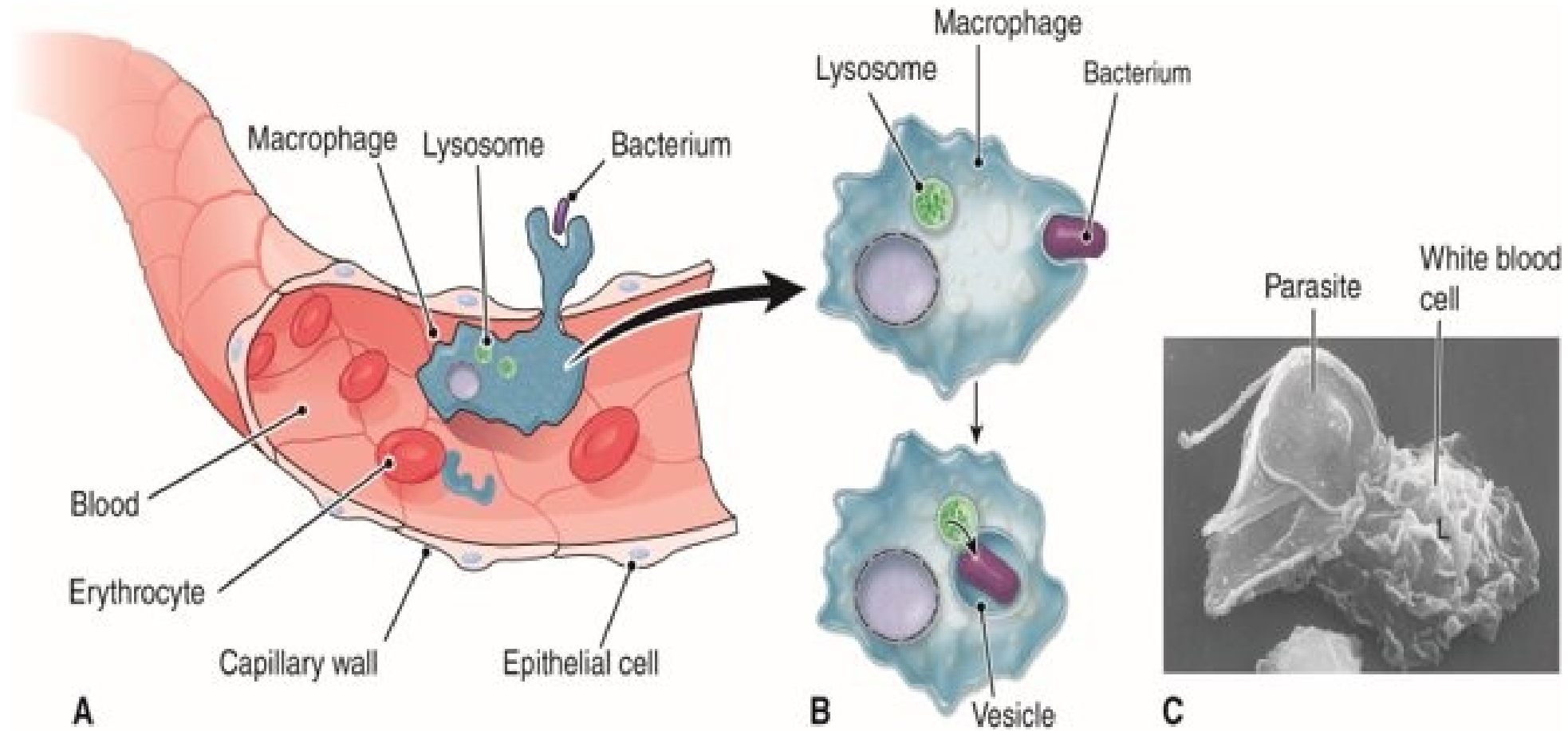
B. Are attracted to
chemicals

3. Positive
Chemotaxis

C. Engulf

4. Phagocytic

D. Leave the bloodstream



<https://www.youtube.com/watch?v=iZYLeIjwe4w>

<https://thewonderofscience.com/phenomenon/2018/7/8/white-blood-cell-chases-bacteria>

Leukocytes -matching

1. Neutrophils

A. Allergies; worms; counteract inflammation

2. Basophils

B. Orchestrate a **specific** pathogen defense (B, T and NK cells)

3. Eosinophils

C. Engulf bacteria; circulating

4. Monocytes

D. Histamine and Heparin

5. Lymphocytes

E. Macrophages; phagocytosis; largest

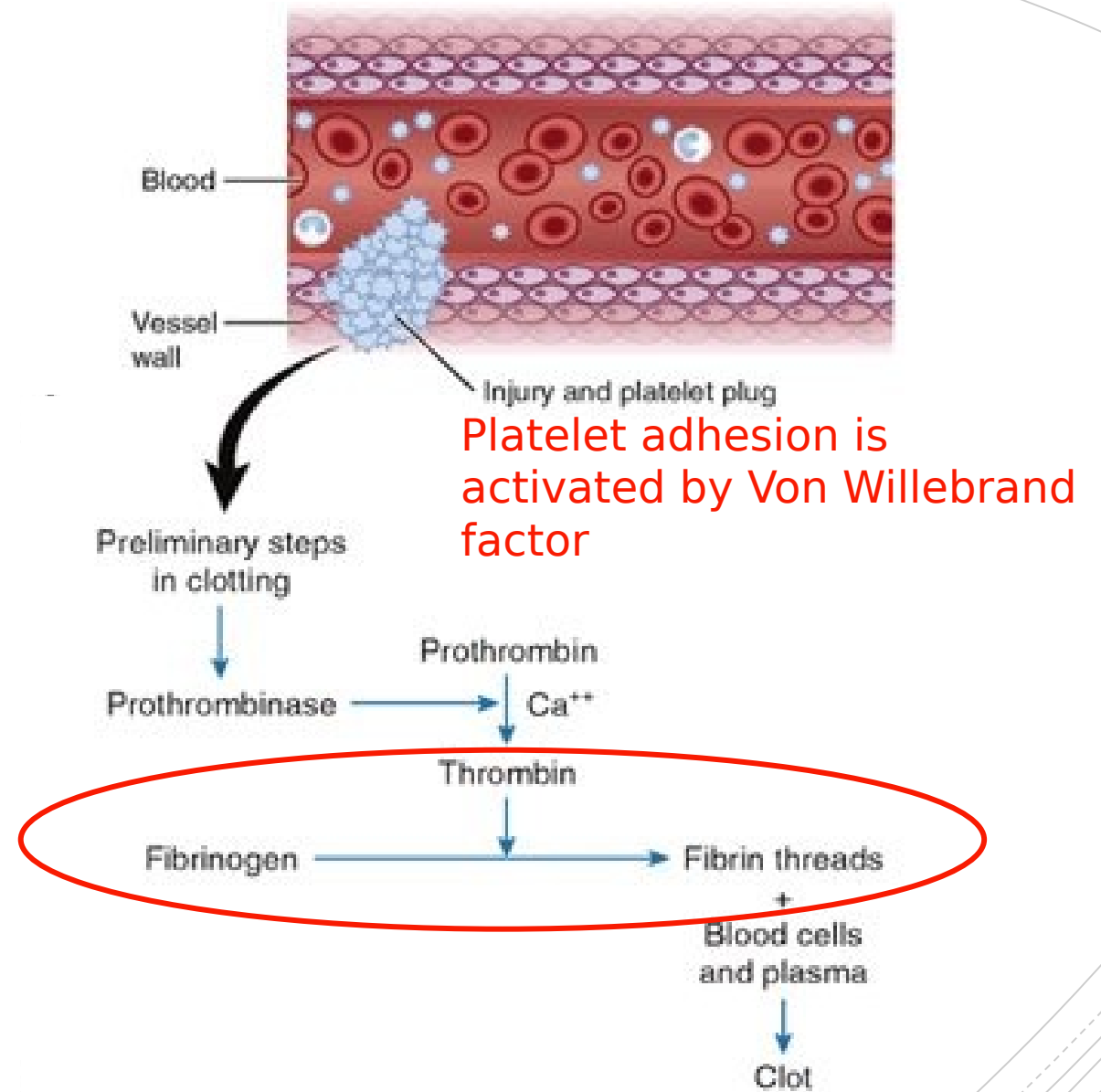
MUST have both platelets and fibrin!

First step: Vasospasm

2 - platelet plug

3- coagulation cascade

Hemostasis



Clotting Cascade – POSITIVE feedback

the nitty-gritty – nice to know not need to know:

EXTRINSIC PATHWAY

(injured cells)

STAT path – quick

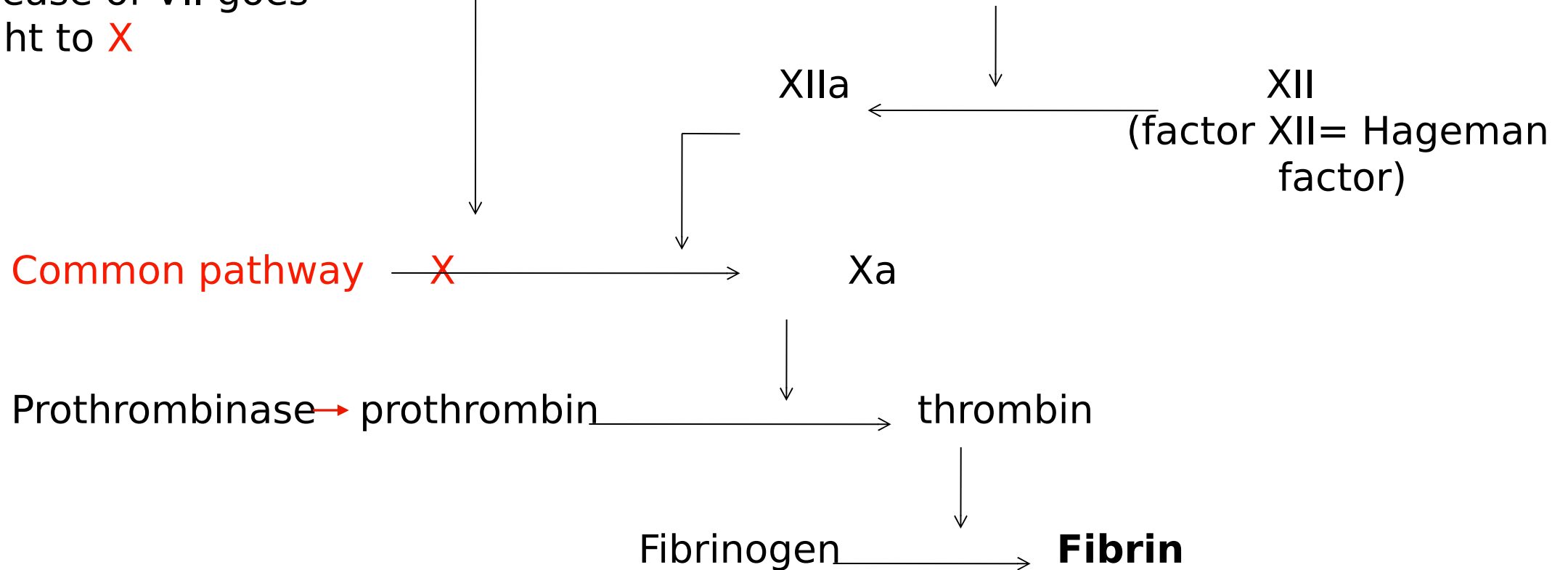
Release of VII goes

Right to **X**

INTRINSIC PATHWAY

(exposure to activators)

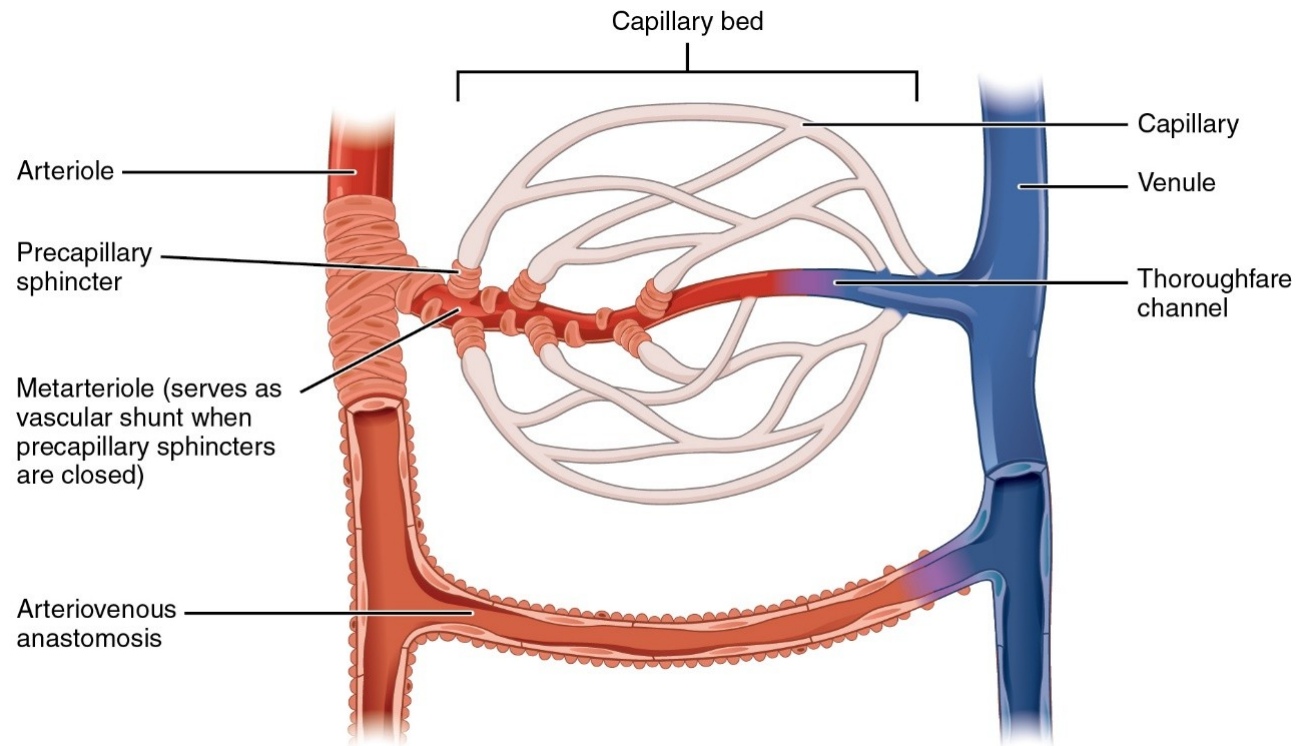
slow – longer time



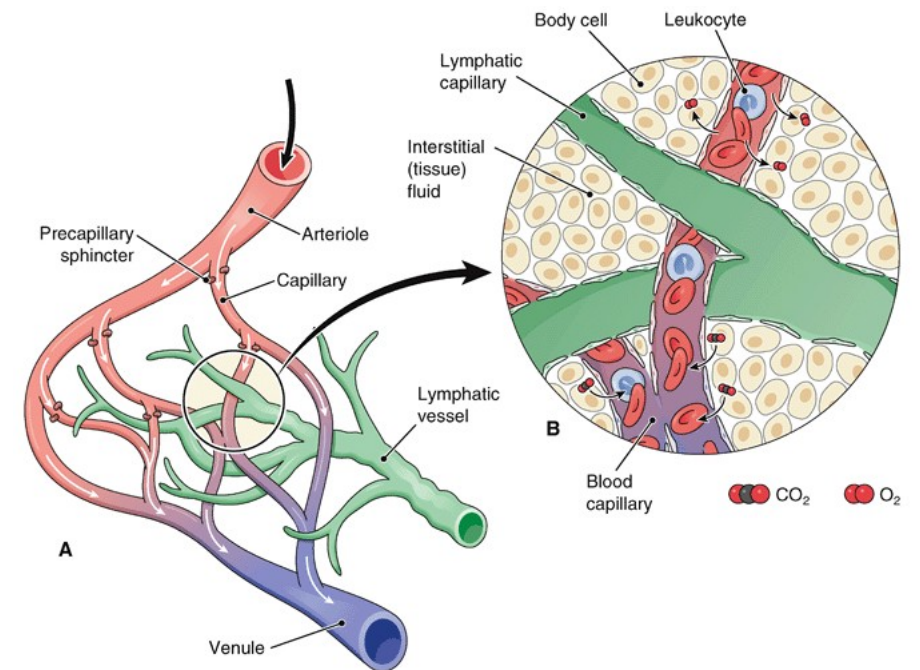
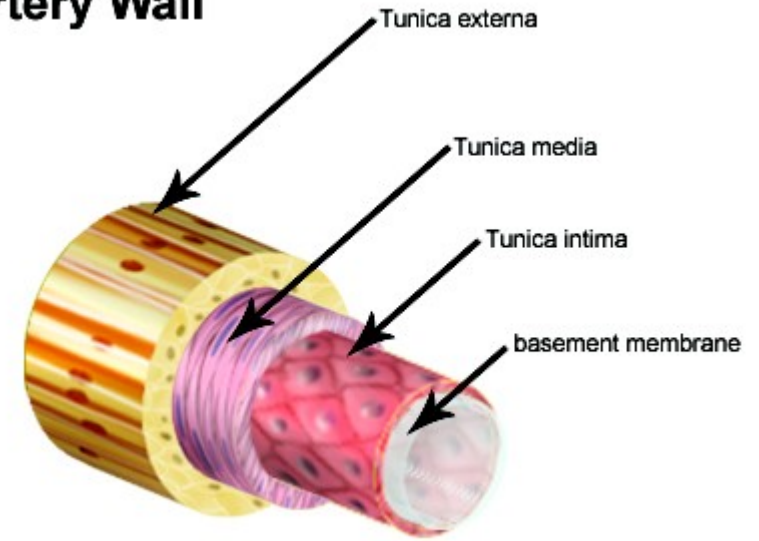
Practice Question

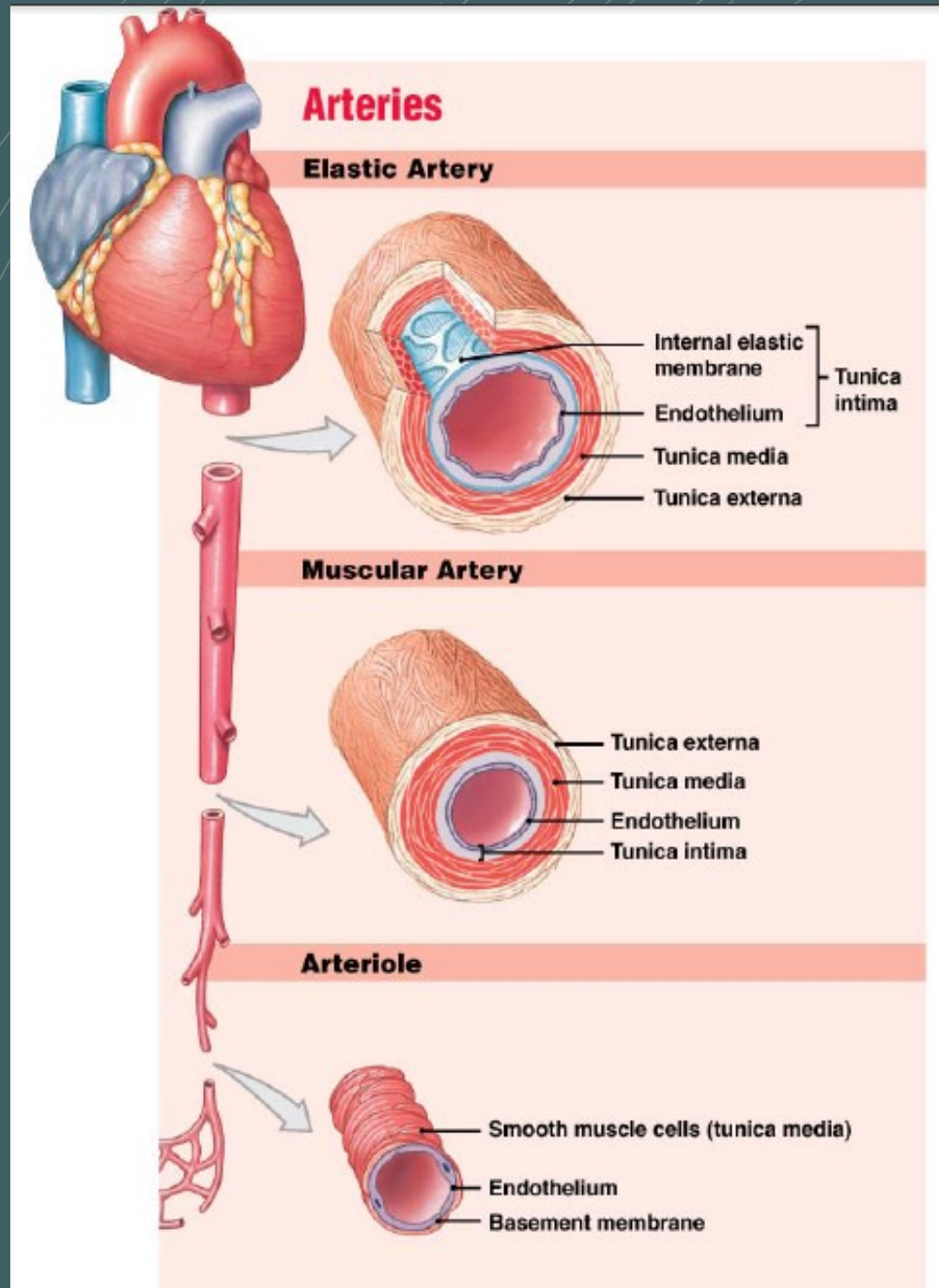
- A person is born without the ability to produce thrombin. Which is true?
- A – they will not be able to synthesize prothrombinase
- B – they will not be able to convert fibrinogen to fibrin
- C – they will still be able to accomplish a successful blood clot
- D – there will be no platelet aggregation

Arteries, Veins, Capillaries



Artery Wall





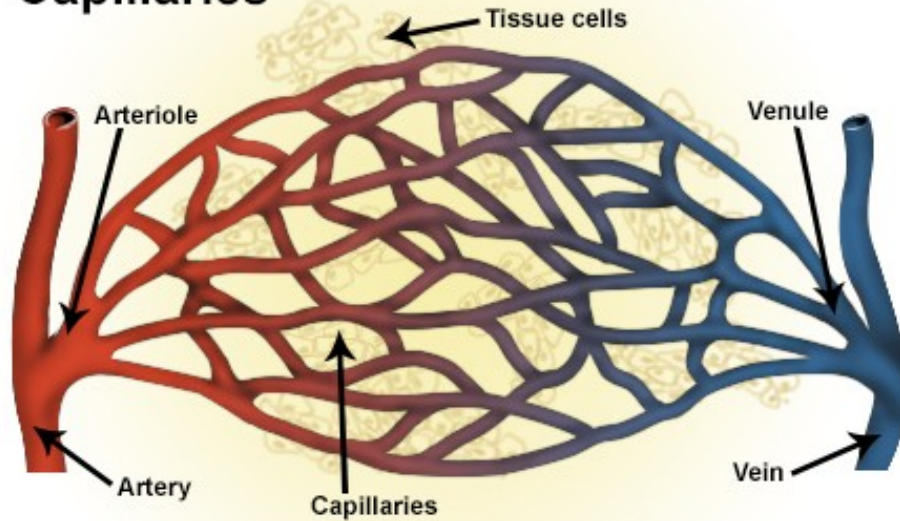
Tunica Media = smooth muscle

This contracts to cause vasoconstriction which will raise blood pressure.

This relaxes to cause vasodilation which lowers blood pressure.

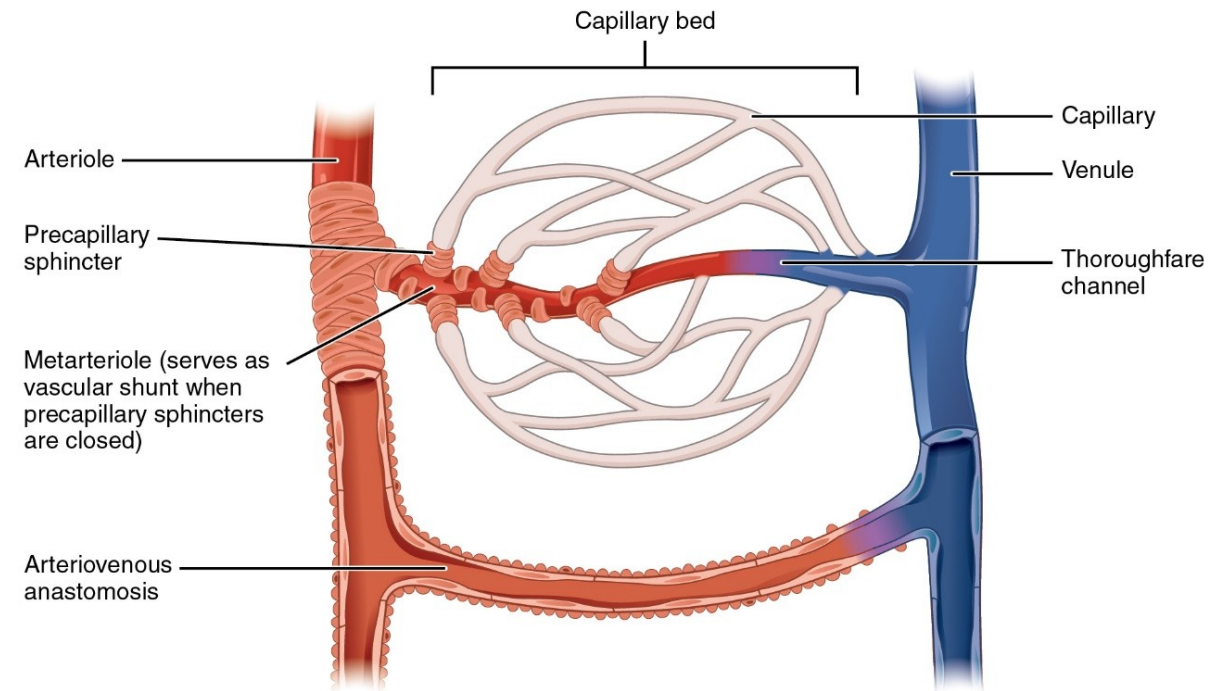
Vessels can autoregulate = adjust size to meet the needs of tissues.

Capillaries

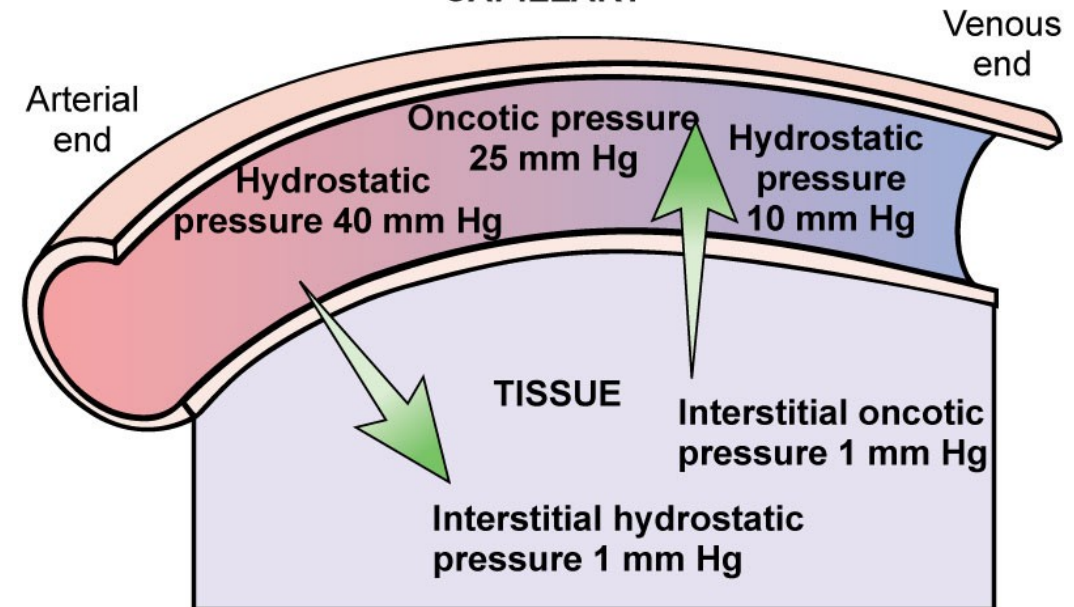


Capillaries are where oxygen and nutrients move from blood to tissues.

Hydrostatic pressure pushes fluid out of the arteriole side and albumin creates oncotic pressure which pulls it back into the venule side (full of waste like CO₂)



CAPILLARY



Reminders:

You can avoid feeling overwhelmed on Sunday by completing something each day – this also leads to better retention of knowledge.

DUE WEDNESDAY: 1st discussion post

DUE SUNDAY:

- Pre-lesson checkpoint
- Module 9 quiz (Respondus)
- Peer response in discussion
- Lab quiz (Respondus)
- EXAM 4 opens MONDAY – covers modules 7 through 9

Struggling?

Let's meet – send me an email and we can arrange a meeting time!